

Comment on “On Gaussian sampling, smoothing parameter and application to lattice signatures”

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We show the key ideas of the above-referenced work for lattice Gaussian sampling [1], which won the best paper award of Asiacrypt 2023, are not new; the same ideas have been proposed by Ling *et al.* in 2014 [2, 3], which are not cited in [1] however.

Next, let us compare the main technical contributions of [1] with [2, 3].

1 “Algebraic Extensions and Sampling” [1, Section 3]

The authors of [1] consider lattice Λ and its sublattices Λ' , and show that if one can sample efficiently in Λ' and Λ/Λ' , then one has a sampler for Λ' , for sampling width above the smoothing parameter of Λ' .

However, the same problem has been studied in [3]. The second part of the following lemma gives such a sampler:

Lemma 1. ([3, Lemma 5]) *Let $\Lambda' \subset \Lambda$ be a pair of nested lattices such that $\epsilon_{\Lambda'}(\sigma) < \frac{1}{2}$. Then*

$$\mathbb{V}(D_{\Lambda, \sigma, \mathbf{c}} \bmod \Lambda', U(\Lambda/\Lambda')) \leq 4\epsilon_{\Lambda'}(\sigma),$$

where $U(\Lambda/\Lambda')$ denotes the uniform distribution over the finite set Λ/Λ' . Conversely, if $\mathbf{L}$ is uniformly distributed in $[\Lambda/\Lambda']$ and $\mathbf{L}'$ is sampled from $D_{\Lambda', \sigma, \mathbf{c}-\mathbf{L}}$, then the distribution $D_{\mathbf{L}+\mathbf{L}'}$ satisfies

$$\mathbb{V}(D_{\mathbf{L}+\mathbf{L}'}, D_{\Lambda, \sigma, \mathbf{c}}) \leq \frac{2\epsilon_{\Lambda'}(\sigma)}{1 - \epsilon_{\Lambda'}(\sigma)}.$$

This lemma has been used to construct a Gaussian sampler in [3, Section V], with the following details given [3, p.6410]:

“We note that the distribution of Alice’s signal can be approximated by D_{Λ_b, σ_s} , when the confidential message is uniformly distributed. More precisely, since Alice transmits $\mathbf{x} \sim D_{\Lambda_e + \lambda_m, \sigma_s}$, by Lemma 1, the density p_{X^n} of $\mathbf{x}$ is close to the discrete Gaussian distribution over Λ_b , if $\lambda_m \in \Lambda_b/\Lambda_e$ is uniformly distributed. In fact, we have $\mathbb{V}(p_{X^n}, D_{\Lambda_b, \sigma_s}) \leq \frac{2\varepsilon}{1-\varepsilon}$ when $\varepsilon \triangleq \epsilon_{\Lambda_e}(\sigma_s) < \frac{1}{2}$.”

The sampler in [1, Section 3] follows the same approach. It is more general, allows a slightly non-uniform distribution on $[\Lambda/\Lambda']$, and refers to a short exact sequence

$$0 \rightarrow \Lambda' \rightarrow \Lambda \rightarrow \Lambda/\Lambda' \rightarrow 0$$

which is just another way to say $\Lambda' \subset \Lambda$. In fact, [1, Lemma 3] and its proof are almost the same as [3, Lemma 5, p.6405] and its proof [3, p.6414], despite being more general. The bound $\frac{\epsilon}{1-\epsilon}$ on the total variational distance of [1, Lemma 3] differs from the $\frac{2\epsilon_{\Lambda'}(\sigma)}{1-\epsilon_{\Lambda'}(\sigma)}$ of [3, Lemma 5, p.6405] by a factor of 2, only due to a slight difference in the definitions of the total variational distance.

Moreover, the sampler of [1, Corollary 3] assumes a uniform distribution over $[\Lambda/\Lambda']$, which is exactly the setting of [3, Lemma 5].

2 “Sampling in Remarkable Lattices” [1, Section 6]

The authors of [1] build efficient samplers for the A_n, D_n, E_n and other lattices.

However, [2, Section IV] has already presented a sampler for Construction A in which codewords are sampled uniformly at random and proved that the resultant distribution is close to a lattice Gaussian if s exceeds the smoothing parameter of $q\mathbb{Z}$.

It further exemplified the method by considering the checkerboard lattice D_n and Gosset lattice E_8 .

3 “A Filtration Sampler” [1, Section 4.2]

A ‘filtration sampler’ is given in [1, Section 4.2], where $\Lambda_0 \subset \Lambda_1 \subset \Lambda_2 \subset \dots$. It is stated that “the smoothing parameter is dominated by the biggest term appearing in the splitting.”

However, the same problem has been considered in [2, Section IV]:

“*Remark 7:* It is straightforward to extend the encoding procedure to multi-level lattices [19]. In this case, the variational distance bound (27) is determined by the flatness factor of the bottom lattice.”

4 “An Exact Calculation of the Smoothing Parameter” [1, p.84]

A method is given in [1, p.84] to find the smoothing parameter using the relation between the lattice Gaussian distribution and the theta series. It is stated that “determining the smoothing parameter of a lattice amounts to find q such that $\Theta_{\Lambda^\vee}(q) - 1 = \epsilon$ ”, where $\Lambda^\vee$ denotes the dual lattice.

However, such a method has already been proposed in [3]:

Corollary 1. ([3, Corollary 1]) *Alternatively, the flatness factor can be expressed on the dual lattice Λ^* as*

$$\epsilon_\Lambda(\sigma) = \Theta_{\Lambda^*}(2\pi\sigma^2) - 1. \quad (1)$$

After a change of variables, (1) is identical to $\Theta_{\Lambda^\vee}(q) - 1 = \epsilon$ in [1, p.84].

5 Conclusion

Reference [1] makes many extensions, but the key ideas are the same as [2, 3] published in IEEE Transactions on Information Theory 9 years before.

References

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